TUTORIAL 7

Linear Dependence, Bases, Dimension, and Row and Column Spaces

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Test linear dependence and independence, find spanning sets, and distinguish the union, intersection, and sum of subspaces. State the scalar field in every test and connect the equations with their geometric interpretation.

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7.1 Linear Dependence and Independence

DEFINITION 7.1

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$.

1. The set S is **linearly dependent** if there are pairwise distinct vectors $v_1, \ldots, v_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that
$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$
2. The set S is **linearly independent** if, for every finite choice of pairwise distinct vectors $v_1, \ldots, v_k \in S$, this equality implies $\alpha_1 = \cdots = \alpha_k = 0$.

The vectors need not be tuples: they may also be polynomials, matrices, or elements of another vector space.

EXAMPLE 7.1

The set
$$\left\{ \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$
is linearly dependent, since
$$1 \cdot \begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 4 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

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Rearranging this equation expresses $\begin{pmatrix} 0 \\ -4 \\ 1 \end{pmatrix}$ in terms of the other vectors.

How Do We Test Dependence?

MOTIVATION AND GUIDING IDEA

- Write a **homogeneous** system whose unknowns are the coefficients in the linear combination.
- Form its matrix and row-reduce.
- **Linear independence:** the only solution is the trivial, or zero, solution.
- **Linear dependence:** there is a nonzero solution. Over an infinite field there are then infinitely many solutions.
- A homogeneous system always has a solution: row reduction cannot produce an inconsistent row.

7.1.1 Useful Criteria for Linear Dependence

1. Every set containing the zero vector is linearly dependent.
2. A singleton consisting of a nonzero vector is linearly independent.
3. **Testing a set for linear dependence.**
   - As in the previous tutorial, set a linear combination equal to zero, obtain a homogeneous system, row-reduce, and determine its solutions.
   - Recall: only the zero solution means independence; a nonzero solution means dependence.
   - Another method from the lecture is to place the vectors as the **rows** of a matrix and row-reduce. If a zero row occurs, the original rows are dependent; if there are no zero rows, they are independent.

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EXERCISE **7.1**

Let

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (0, 2, 1)^T, \qquad v_3 = (3, -1, -2)^T.$$

Please determine linear dependence, please find all dependence relations, and please express one vector using the others.

WORKED SOLUTION

Write $av_1 + bv_2 + cv_3 = 0$. Its coordinates give

$$a + 3c = 0, \qquad a + 2b - c = 0, \qquad b - 2c = 0.$$

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EXERCISE 7.1 CONTINUED

Thus $a = -3c$, $b = 2c$, and the middle equation is $-3c + 4c - c = 0$. All relations are

$$(a, b, c) = t(-3, 2, 1), \qquad t \in \mathbb{R}.$$

In particular, $-3v_1 + 2v_2 + v_3 = 0$, so $v_3 = 3v_1 - 2v_2$. This is a nontrivial dependence. In $av_1 + bv_2 = 0$, the first and third coordinates force $a = b = 0$, so the first two vectors are independent.

7.2 Supplementary Exercises: Span, Redundancy, and the Scalar Field

EXERCISE 7.2

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + z = 0\}, \qquad u = (2, -1, 0), \quad v = (0, 1, -2), \quad w = (2, 0, -2).$$

Please prove $U = \text{span}\{u, v\}$. Please test independence and the effect of adding w. Please determine whether $x + 2y + z = 3$ defines a subspace.

WORKED SOLUTION

$$au + bv = (2a, -a + b, -2b), \qquad 2a + 2(-a + b) - 2b = 0.$$

Thus $\text{span}\{u, v\} \subseteq U$. Conversely, $(x, y, z) \in U$ implies $y = -(x + z)/2$, so

$$(x, y, z) = \frac{x}{2}u - \frac{z}{2}v.$$

Hence $U = \text{span}\{u, v\}$. In $au + bv = 0$, the first and third coordinates give $a = b = 0$, proving independence. Since $w = u + v$, adding w preserves the span but introduces the dependence $u + v - w = 0$. The set defined by $x + 2y + z = 3$ is not a subspace because it excludes the zero vector.

EXERCISE 7.3

For $S = \{1, 2 + i\} \subseteq \mathbb{C}$, please test linear dependence and spanning over $\mathbb{R}$ and over $\mathbb{C}$.

WORKED SOLUTION

Over $\mathbb{R}$, $a + b(2 + i) = 0$ gives $b = 0$ from the imaginary part, then $a = 0$. Thus the pair is independent. Every $x + iy \in \mathbb{C}$ satisfies

$$x + iy = (x - 2y) \cdot 1 + y(2 + i),$$

so the pair spans $\mathbb{C}$ over $\mathbb{R}$. Over $\mathbb{C}$, the nontrivial relation

$$-(2 + i) \cdot 1 + 1 \cdot (2 + i) = 0$$

shows dependence. The pair still spans because 1 alone spans $\mathbb{C}$ over itself.

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EXERCISE 7.4 FOUNDATION

Work over $\mathbb{R}$. Let $u = (1, 1, 0)^T$ and $v = (0, 1, 1)^T$.

1. Please describe $U = \text{span}\{u, v\}$ by an equation.
2. Please determine whether $(3, 7, 4)^T$ and $(3, 6, 4)^T$ belong to U, finding coefficients when possible.
3. Please determine whether u, v are independent.

WORKED SOLUTION

For $a, b \in \mathbb{R}$,

$$au + bv = (a, a + b, b)^T.$$

Every vector in the span therefore satisfies $y = x+z$. Conversely, if $(x, y, z)^T$ satisfies $y = x+z$, then

$$(x, y, z)^T = xu + zv.$$

Thus $U = \{(x, y, z)^T : y = x + z\}$. For the first vector, $7 = 3 + 4$, and the vector is $3u + 4v$. For the second, $6 \neq 3 + 4$, so it is not in the span. Finally, if $au + bv = 0$, the first and third coordinates give $a = b = 0$. Thus u, v are independent; in particular, the coefficients found above are unique.

MOTIVATION AND GUIDING IDEA

Motivation. A homogeneous linear constraint allows two admissible states to be added, or an admissible state to be scaled, without leaving the admissible set. Span describes which states can be constructed; independence identifies components that are not redundant. Together, these ideas describe a system using a small collection of independent components.

NOTATION AND CONVENTIONS

Notation guide. $\mathbb{F}$ is the scalar field, V a vector space over it, 0_V the zero vector, and $\alpha \in \mathbb{F}$ a scalar. $v \in S$ denotes membership and $S \subseteq V$ inclusion. $\text{span}_{\mathbb{F}} S$ is the span over the indicated field. $U \cup W$ denotes union, $U \cap W$ intersection, $U + W$ a sum of subspaces, and $U \oplus W$ a direct sum. The symbol $\setminus$ denotes set difference and $\varnothing$ the empty set.

DEFINITION 7.2

A precise formulation with hypotheses. For $S \subseteq V$, its span consists of all finite sums $\sum_{j=1}^{k} \alpha_j v_j$ with $v_j \in S$ and $\alpha_j \in \mathbb{F}$, including the empty sum, which equals 0_V. It is the smallest subspace containing S. Independence means that every finite zero linear combination of distinct members of S has only zero coefficients. In particular, the empty set is independent and its span is $\{0_V\}$.
For subspaces $U, W \leq V$ over the same field, $U + W$ is the smallest subspace containing both. The condition $U \cap W = \{0_V\}$ is equivalent to every $x \in U + W$ having a unique expression $x = u + w$ with $u \in U, w \in W$. Indeed, two expressions give $u - u' = w' - w \in U \cap W$. Conversely, if $z \in U \cap W$, then $z = z + 0_V = 0_V + z$, so uniqueness forces $z = 0_V$.

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NOTE

A vector is an element of the vector space, so a matrix or polynomial can also be a vector. In an independence test, the unknowns are the *coefficients of the combination*; in a span-membership test, the target vector is given. For a nonhomogeneous condition such as $x + 2y + z = 3$, first check whether the zero vector belongs to the set.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Compute with matrices while checking dimensions and multiplication order, recognize symmetric structure, and choose bases. Connect algebraic calculations with sums and intersections of subspaces.

7.3 Bases and Dimension

DEFINITION 7.3

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 7.2

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d + 1.$$

For example, the standard basis of $\mathbb{R}^3$ is

$$e_1 = (1, 0, 0)^T;\ e_2 = (0, 1, 0)^T;\ e_3 = (0, 0, 1)^T.$$

The standard basis of $\mathbb{R}_3[x]$ is $(1, x, x^2, x^3)$.

RECALL AND TOOLS

In a vector space of known finite dimension n, any two of the following three properties suffice to establish that a set is a basis:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

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EXERCISE 7.5

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, i)^T, \quad v_2 = (2 - i, 1, 0)^T, \quad v_3 = (4 - 3i, 2 - i, -1 + i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(1 + i)v_1 + (2 - i)v_2 = v_3.$$

$$(1 + i) + (2 - i)^2 = 4 - 3i, \qquad (1 + i)i = -1 + i.$$

Thus $(1+i)v_1 + (2-i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

EXERCISE 7.6

Let $\{u, v, w\}$ be a basis of $\mathbb{R}^3$. Please determine which sets are also bases.

1. $\{u, u + 2v, u - 2v\}$.
2. $\{u/3, v/5, w/7\}$.
3. $\{u + v + w, u + 3v, 2v - w\}$.

WORKED SOLUTION

Each set has three vectors in a space of dimension 3, so independence suffices.

1. No: $-2u + (u + 2v) + (u - 2v) = 0$ is a nontrivial dependence.
2. Yes. If $a(u/3)+b(v/5)+c(w/7) = 0$, independence of u, v, w gives $a/3 = b/5 = c/7 = 0$, hence $a = b = c = 0$.
3. No: $(u + v + w) - (u + 3v) + (2v - w) = 0$ is a nontrivial dependence.

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7.4 The Dimension Formula

PROPOSITION 7.1

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

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PROPOSITION 7.2

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 7.7

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n - 1$ and $W \nsubseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2 + 4\}$, $W = \{p \in \mathbb{R}_2[X] : p'(2) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

   $$n - 1 = \dim U < \dim(U + W) \leq n.$$

   Thus $\dim(U + W) = n$ and $U + W = V$. The dimension formula gives

   $$n = (n - 1) + \dim W - \dim(U \cap W),$$

   hence $\dim(U \cap W) = \dim W - 1$.

2. The generators of U are independent, so $\dim U = 2$. Also $1 \in W \setminus U$. For W,

   $$p(X) = aX^2 + bX + c \in W \iff 4a + b = 0 \iff b = -4a.$$

   $$W = \operatorname{span}\{1, X^2 - 4X\}, \qquad \dim W = 2.$$

   Part 1 gives $U + W = V$. The sum is not direct, since

   $$(X - 2)^2 = (X^2 + 4) - 4X \in U, \qquad \frac{d}{dX}(X - 2)^2\Big|_{X=2} = 0.$$

   $$U \cap W = \operatorname{span}\{(X - 2)^2\}, \quad \dim(U \cap W) = 1, \quad \dim(U + W) = 2 + 2 - 1 = 3.$$

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7.5 Bases and Dimension

DEFINITION 7.4

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

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EXAMPLE 7.3

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d+1.$$

For example, the standard basis of $\mathbb{R}^4$ is

$$e_1 = (1,0,0,0)^T;\ e_2 = (0,1,0,0)^T;\ e_3 = (0,0,1,0)^T;\ e_4 = (0,0,0,1)^T.$$

The standard basis of $\mathbb{R}_1[x]$ is $(1, x)$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \text{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

PROPOSITION 7.3

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

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RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 7.8

Please determine whether these vectors form a basis of $\mathbb{C}^3$. If not, please give a dependence relation.

$$v_1 = (1, 0, 2i)^T, \quad v_2 = (1 - i, 1, 0)^T, \quad v_3 = (5 + 2i, 1 + 2i, -2 + 4i)^T.$$

WORKED SOLUTION

The first two vectors are independent: in $av_1 + bv_2 = 0$, the second coordinate gives $b = 0$, and then the first gives $a = 0$. Computing coordinates gives

$$(2 + i)v_1 + (1 + 2i)v_2 = v_3.$$

$$(2 + i) + (1 + 2i)(1 - i) = 5 + 2i, \qquad (2 + i)2i = -2 + 4i.$$

Thus $(2 + i)v_1 + (1 + 2i)v_2 - v_3 = 0$ is nontrivial. The rank is 2, so the triple is not a basis of $\mathbb{C}^3$.

EXERCISE 7.9

Work in $M_3(\mathbb{R})$. The matrix unit E_{ij} has entry 1 at (i, j) and zero elsewhere. Define

$$D_i = E_{ii} \quad (1 \le i \le 3), \qquad S_{ij} = E_{ij} + E_{ji}, \quad K_{ij} = E_{ij} - E_{ji} \quad (1 \le i < j \le 3).$$

$$\mathcal{B}_s = \{D_1, D_2, D_3, S_{12}, S_{13}, S_{23}\}, \qquad \mathcal{B}_a = \{K_{12}, K_{13}, K_{23}\}.$$

Please prove that $\mathcal{B}_s$ is a basis of the symmetric matrices and $\mathcal{B}_a$ is a basis of the skew-symmetric matrices. Please prove that their union is a basis of $M_3(\mathbb{R})$ and please compute the three dimensions.

WORKED SOLUTION

We have $D_i^T = D_i$, $S_{ij}^T = S_{ij}$, and $K_{ij}^T = -K_{ij}$. Every symmetric $A = (a_{ij})$ has the representation

$$A = \sum_{i=1}^{3} a_{ii} D_i + \sum_{1 \le i < j \le 3} a_{ij} S_{ij}.$$

The diagonal and upper-diagonal entries determine all coefficients uniquely. Thus the family spans and is independent, giving a basis of size $3 + 3 = 6$.

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EXERCISE 7.9 CONTINUED

For skew-symmetric A, the diagonal is zero and $a_{ji} = -a_{ij}$. Hence

$$A = \sum_{1 \le i < j \le 3} a_{ij} K_{ij}.$$

Again, the entries above the diagonal uniquely determine the coefficients, so $\mathcal{B}_a$ is a basis of size 3.

Every $A \in M_3(\mathbb{R})$ satisfies

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric and the second skew-symmetric, so the union spans all matrices. The intersection of the two subspaces is $\{0\}$: $A^T = A = -A$ gives $2A = 0$, hence $A = 0$. Any zero combination of the union therefore splits into a zero combination in each basis, forcing every coefficient to vanish. The union is a basis, with dimensions

$$\dim \mathrm{Sym}_3(\mathbb{R}) = 6, \quad \dim \mathrm{ASym}_3(\mathbb{R}) = 3, \quad \dim M_3(\mathbb{R}) = 9 = 6 + 3.$$

7.6 The Dimension Formula

PROPOSITION 7.4

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \le \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

PROPOSITION 7.5

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 7.10

1. Let V have dimension n, and let U, W be subspaces with $\dim U = n - 1$ and $W \not\subseteq U$. Please prove $V = U + W$ and $\dim(U \cap W) = \dim W - 1$.
2. $V = \mathbb{R}_2[X]$, $U = \mathrm{span}\{X, X^2 + 9\}$, $W = \{p \in \mathbb{R}_2[X] : p'(-3) = 0\}$. Please compute the dimensions of the sum and intersection, and please determine whether the sum equals V and is direct.

WORKED SOLUTION

1. Choose $w \in W \setminus U$. Then $U \subsetneq U + W \subseteq V$, so

$$n - 1 = \dim U < \dim(U + W) \le n.$$

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EXERCISE 7.10 CONTINUED

Thus $\dim(U+W)=n$ and $U+W=V$. The dimension formula gives

$$n=(n-1)+\dim W-\dim(U\cap W),$$

hence $\dim(U\cap W)=\dim W-1$.

2. The generators of U are independent, so $\dim U=2$. Also $1\in W\setminus U$. For W,

$$p(X)=aX^2+bX+c\in W \iff -6a+b=0 \iff b=6a.$$

$$W=\operatorname{span}\{1,X^2+6X\},\qquad \dim W=2.$$

Part 1 gives $U+W=V$. The sum is not direct, since

$$(X+3)^2=(X^2+9)+6X\in U,\qquad \frac{d}{dX}(X+3)^2\Big|_{X=-3}=0.$$

$$U\cap W=\operatorname{span}\{(X+3)^2\},\quad \dim(U\cap W)=1,\quad \dim(U+W)=2+2-1=3.$$

MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax=b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A)=\operatorname{Row}(A)\subseteq\mathbb{F}^n$ is the row space; $\mathcal{C}(A)=\operatorname{Col}(A)\subseteq\mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A)=\ker A=\{x\in\mathbb{F}^n : Ax=0\}$ is the kernel; $\operatorname{nullity}(A)=\dim\ker A$ is the nullity; $\operatorname{rank}(A)=\dim\operatorname{Col}(A)=\dim\operatorname{Row}(A)$ is the rank.
- $\operatorname{span} S$ consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.

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- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

DEFINITION 7.5

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\operatorname{Im} T_A = \operatorname{Col}(A)$ and $\ker T_A = N(A)$, so

$$\operatorname{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \operatorname{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

7.7 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 7.6

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

PROPOSITION 7.7

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

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RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

EXERCISE 7.11

$$A = \begin{pmatrix} 2 & 6 & -4 \\ 1 & 3 & -2 \end{pmatrix}.$$

Please find a basis and dimension for the row space and column space of A.

WORKED SOLUTION

Swap the rows and subtract twice the first row from the second. The column-space basis is taken from the original matrix.

$$A \sim \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & -2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\text{Row}(A) = \text{span}\{(1, 3, -2)\}, \qquad \text{Col}(A) = \text{span}\{(2, 1)^T\}.$$

$$\text{Col}_2(A) = 3\,\text{Col}_1(A), \qquad \text{Col}_3(A) = -2\,\text{Col}_1(A), \qquad \dim \text{Row}(A) = \dim \text{Col}(A) = 1.$$

EXERCISE 7.12

Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Please find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\text{rank}(AB) = \text{rank}(I_m) = m \leq \text{rank}(A) \leq m$.
Thus $\text{rank}(A) = m$. For B, $m = \text{rank}(AB) \leq \text{rank}(B) \leq m$, so $\text{rank}(B) = m$ as well.
Since $\text{rank}(A) = m \leq \min(m, n)$, we obtain $m \leq n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

7.7.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

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Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.
   - Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
   - Its nonzero echelon rows form a basis.
   - Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

MOTIVATION AND GUIDING IDEA

- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE **7.13**

$$S = \{(1,2,0,1,-1)^T, (2,4,1,3,0)^T, (1,2,1,2,2)^T\} \subseteq \mathbb{R}^5.$$

Please prove that S is linearly independent and please extend it to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

The three nonzero echelon rows prove independence. Pivot columns are $1, 3, 5$. Adding e_2, e_4 gives five independent rows: the pivot columns first force the three echelon-row coefficients to vanish, and columns $2, 4$ then force the remaining two coefficients to vanish.

$$\begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 4 & 1 & 3 & 0 \\ 1 & 2 & 1 & 2 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$B = S \cup \{e_2, e_4\}.$$

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EXERCISE 7.14

$$V = \text{span}\{(1,3,2,-1)^T, (2,7,5,0)^T, (0,1,1,2)^T\} \subseteq \mathbb{R}^4.$$

Please find a basis and dimension for V and please extend that basis to a basis of $\mathbb{R}^4$.

WORKED SOLUTION

The first two coordinates show that the two vectors in B are independent. They span all three given vectors. The final determinant verifies the extension.

$$(2,7,5,0)^T = 2(1,3,2,-1)^T + (0,1,1,2)^T.$$

$$B = \{(1,3,2,-1)^T, (0,1,1,2)^T\}, \qquad \dim V = 2.$$

$$\det \begin{pmatrix} 1 & 3 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1, \qquad D = B \cup \{e_3, e_4\}.$$

7.8 The null space and nullity of a matrix

DEFINITION 7.6

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \text{null}(A).$$

PROPOSITION 7.8

For $A \in \mathbb{F}^{m\times n}$,

$$\text{rank}(A) + \text{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\text{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\text{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \text{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \text{null}(A) = n - \text{rank}(A).$$

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- Therefore,

$$\operatorname{rank}(A) + \operatorname{null}(A) = \operatorname{rank}(A) + \big(n - \operatorname{rank}(A)\big) = n.$$

EXERCISE 7.15

$$A = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 2 \end{pmatrix}.$$

Please find a basis and dimension for the kernel, please give every solution of $Ax = 0$, and please check rank-nullity.

WORKED SOLUTION

The first two columns are pivot columns. Every solution arises from $t \in \mathbb{R}$; the nonzero spanning vector is a kernel basis.

$$x_1 - 3x_2 = 0, \quad x_2 + 2x_3 = 0, \quad x_3 = t \implies x_2 = -2t,\ x_1 = -6t.$$

$$\ker A = \operatorname{span}\{(-6, -2, 1)^T\}, \qquad \operatorname{rank} A + \dim \ker A = 2 + 1 = 3.$$

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

7.9 Interpreting matrix-vector multiplication

EXERCISE 7.16

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n\times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Please show that

$$J = \operatorname{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.
Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

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EXERCISE 7.16 CONTINUED

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \ldots, b_m$:

   $$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

   $$y \in \mathrm{span}\{b_1, \ldots, b_m\} = \mathrm{Col}(B).$$

   This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

   Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \ldots, \eta_m \in \mathbb{F}$ such that

   $$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

   Define

   $$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

   The preceding calculation gives

   $$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

   Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

   $$w \in J.$$

   This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

EXERCISE 7.17

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n \times n}$ satisfy

$$AB = 0.$$

Please show that

$$\mathrm{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

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EXERCISE 7.17 CONTINUED

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \mathrm{Col}(B)$.
By the definition of column space, $v \in \mathrm{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

and therefore

$$Av = 0.$$

We have proved that $Av = 0$ for every $v \in \mathrm{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\mathrm{Col}(B) \subseteq N(A),$$

as required.

7.10 Exercises

EXERCISE 7.18

$$M = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & -1 & 5 \end{pmatrix}, \qquad y = (4, -1, 3, 10)^T.$$

Please find row-space and column-space bases and the number of dependent rows. Please decide whether $Mx = y$ is solvable and whether an existing solution is unique. Please replace the fourth entry of y by 9 and please find every solution.

WORKED SOLUTION

The first two rows are independent and the other two depend on them. The third column is twice the first minus the second, so the listed columns form a basis. The row relations give necessary right-hand-side conditions; they are sufficient because the first two equations can be solved for any two right-hand sides. The original right-hand side is inconsistent. After correction, setting $x_3 = t$ gives the displayed solutions. The one-dimensional kernel prevents uniqueness.

$$R_3 = R_1 + R_2, \qquad R_4 = 2R_1 - R_2, \qquad \operatorname{rank} M = 2.$$

$$B_{\text{row}} = \{(1, 0, 2), (0, 1, -1)\}, \quad B_{\text{col}} = \{(1, 0, 1, 2)^T, (0, 1, 1, -1)^T\}.$$

$$y \in \operatorname{Col} M \iff y_3 = y_1 + y_2, \quad y_4 = 2y_1 - y_2.$$

$$2 \cdot 4 - (-1) = 9 \neq 10, \qquad \ker M = \operatorname{span}\{(-2, 1, 1)^T\}.$$

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EXERCISE 7.18 CONTINUED

$$x = (4,-1,0)^T + t(-2,1,1)^T, \qquad t \in \mathbb{R}.$$

Additional practice

EXERCISE 7.19

$$U = \operatorname{span}\{1, x, 2+x\}, \qquad W = \{p \in \mathbb{R}_3[x] : p(-1) = p(2) = 0\}.$$

Please find bases and dimensions for $U + W$ and $U \cap W$.

WORKED SOLUTION

A polynomial of degree at most one with two distinct roots is zero. Thus the intersection is zero and the sum is direct. The four listed polynomials are independent by their degrees and span the ambient space. The intersection has the empty basis.

$$U = \operatorname{span}\{1, x\}, \qquad p \in W \iff p = (x+1)(x-2)(a+bx).$$

$$W = \operatorname{span}\{x^2 - x - 2,\ x^3 - x^2 - 2x\}, \qquad \dim W = 2.$$

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0, \qquad U + W = \mathbb{R}_3[x], \quad \dim(U+W) = 4.$$

$$B_{U+W} = (1, x, x^2 - x - 2, x^3 - x^2 - 2x).$$

EXERCISE 7.20 FOUNDATION

$$A = \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 1 & -1 & 1 & 5 \end{pmatrix}, \qquad b = (2,4,6)^T.$$

Please find row-space, column-space and kernel bases and please check rank-nullity. Please give a solvability condition for a general right-hand side and please find all solutions for the given b.

WORKED SOLUTION

The first two rows are independent and pivot columns are $1, 3$. The kernel vectors have independent free entries in positions $2, 4$. The row relation gives the solvability condition, and the first two equations prove sufficiency. The displayed solutions are a translate of the kernel.

$$R_3 = R_1 + R_2, \quad B_{\text{row}} = \{(1,-1,0,2), (0,0,1,3)\}, \quad \operatorname{rank} A = 2,$$

$$B_{\text{col}} = \{(1,0,1)^T, (0,1,1)^T\},$$

$$x_2 = s, \quad x_4 = t \quad \Longrightarrow \quad x_1 = s - 2t, \quad x_3 = -3t,$$

$$B_{\text{ker}} = \{(1,1,0,0)^T, (-2,0,-3,1)^T\}, \qquad 2 + 2 = 4.$$

$$\exists x:\ Ax = b \iff b_3 = b_1 + b_2,$$

$$x = (2,0,4,0)^T + s(1,1,0,0)^T + t(-2,0,-3,1)^T, \qquad s, t \in \mathbb{R}.$$

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EXERCISE 7.21 CHALLENGE

Let $A \in M_n(F)$ over any field. Please prove

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) \geq n,$$

with equality if and only if $A^2 = A$.

WORKED SOLUTION

Put $U = \operatorname{Col} A$ and $W = \operatorname{Col}(I_n - A)$. Since $x = Ax + (I_n - A)x$, we have $U + W = F^n$. The dimension formula gives

$$\operatorname{rank} A + \operatorname{rank}(I_n - A) = n + \dim(U \cap W) \geq n.$$

If equality holds, then $U \cap W = \{0\}$. For every x, the vector $A(I_n - A)x = (I_n - A)Ax$ belongs to both spaces, so it is zero. Hence $A^2 = A$. Conversely, assume $A^2 = A$ and let $y = Au = (I_n - A)v$. The first representation gives $Ay = y$, while the second gives $Ay = (A - A^2)v = 0$. Therefore $y = 0$, the intersection is trivial, and equality follows.

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EXERCISE 7.22 CHALLENGE

Let $m, n \geq 2$ and let $W \neq \{0\}$ be a subspace of $M_{m\times n}(F)$ in which every matrix has rank at most 1. Please prove that at least one of the following holds:

1. There is a fixed $u \in F^m \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $v \in F^n$.
2. There is a fixed $v \in F^n \setminus \{0\}$ such that every $A \in W$ has the form uv^T for some $u \in F^m$.

WORKED SOLUTION

Every nonzero rank-one matrix is uv^T with $u, v \neq 0$: choose a nonzero column u and collect the scalar multiples expressing the other columns into v^T. First consider uv^T and rs^T. If both u, r and v, s are independent pairs, the matrix with rows v^T, s^T has rank 2 and maps onto F^2. Thus there are x, y with

$$v^T x = 1,\ s^T x = 0, \qquad v^T y = 0,\ s^T y = 1.$$

The sum sends x to u and y to r, so its rank is at least 2. Consequently a sum of rank at most 1 forces dependence of at least one pair of factors. Choose $A_0 = u_0 v_0^T \in W$ nonzero. If every column space lies in $\text{span}\{u_0\}$, the first alternative holds. Otherwise choose B whose column direction is independent of u_0. Since $A_0 + B \in W$, the preceding argument forces the row factor of B to be a multiple of v_0. Absorbing that nonzero scalar into its column factor gives $B = rv_0^T$ with r, u_0 independent. For any nonzero $C = cd^T \in W$, suppose d, v_0 were independent. Applying the same argument to $A_0 + C$ forces $c \in \text{span}\{u_0\}$; applying it to $B + C$ forces $c \in \text{span}\{r\}$. This gives $c = 0$, a contradiction. Every row factor is therefore a multiple of v_0, proving the second alternative.

MOTIVATION AND GUIDING IDEA

Motivation. Matrix multiplication describes composition of linear operations: in ABx, apply B first and then A. Checking dimensions reveals in advance whether the composition is possible. A basis gives a unique description of every vector, and dimension counts the coefficients required for that description.

NOTATION AND CONVENTIONS

Notation guide. $M_{m\times n}(\mathbb{F})$ is the space of matrices with m rows and n columns; a_{ij} is the entry in row i, column j. $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. e_j is the jth standard basis vector, and Ae_j is column j of A. A^T denotes transpose, $\text{diag}(d_1, \ldots, d_n)$ a diagonal matrix, $|B|$ the cardinality of a finite set, and $\dim_{\mathbb{F}} V$ the dimension over $\mathbb{F}$.

DEFINITION 7.7

Hypotheses to check. If $A \in M_{k\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, then $AB \in M_{k\times n}(\mathbb{F})$ with $(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj}$. Also $(AB)^T = B^T A^T$. Over $\mathbb{C}$, the transpose A^T does not include complex conjugation.

For square matrices, define $\text{Sym}_n(\mathbb{F}) = \{A : A^T = A\}$ and $\text{ASym}_n(\mathbb{F}) = \{A : A^T = -A\}$. Both sets are subspaces over any field. If the characteristic is different from 2, a skew-symmetric matrix has zero diagonal and every matrix has a unique symmetric/skew-symmetric decomposition using division by 2. In characteristic 2 the two conditions coincide

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and do not imply a zero diagonal.
If $\dim_{\mathbb{F}} V = n < \infty$, a set of n vectors is a basis if and only if it is independent, and if and only if it spans. For $U, W \leq V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; subtracting the intersection dimension prevents shared directions from being counted twice.

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

DEFINITION 7.8

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.
The following are equivalent: A is invertible; its reduced row-echelon form is I_n; $\operatorname{rank} A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.
If row operations are applied in the order $E_1, E_2, \ldots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

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EXERCISE 7.23 CHALLENGE

Let $n \geq 2$. Let W consist of all real $n \times n$ matrices whose row sums and column sums all equal the same number. Please prove that W is a subspace and find a basis and its dimension. You may use E_{ij}, the matrix with a single 1 in position (i, j).

WORKED SOLUTION

Addition adds the common sums, scalar multiplication scales them, and zero belongs to W. Hence W is a subspace. Let W_0 be the matrices with all row and column sums zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn}.$$

Each lies in W_0. Every $A \in W_0$ satisfies

$$A = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} a_{ij} F_{ij}.$$

The upper left $(n-1) \times (n-1)$ blocks agree. Zero row sums then determine the last column, zero column sums determine the last row, and the final entry is determined as well. These matrices are independent: entry (i, j) with $i, j < n$ reads off the coefficient of F_{ij}. If the common sum of $A \in W$ is c, then $A - cI_n \in W_0$. Since $I_n \notin W_0$, a basis is

$$\{I_n\} \cup \{F_{ij} : 1 \leq i, j < n\}, \qquad \dim W = 1 + (n-1)^2.$$

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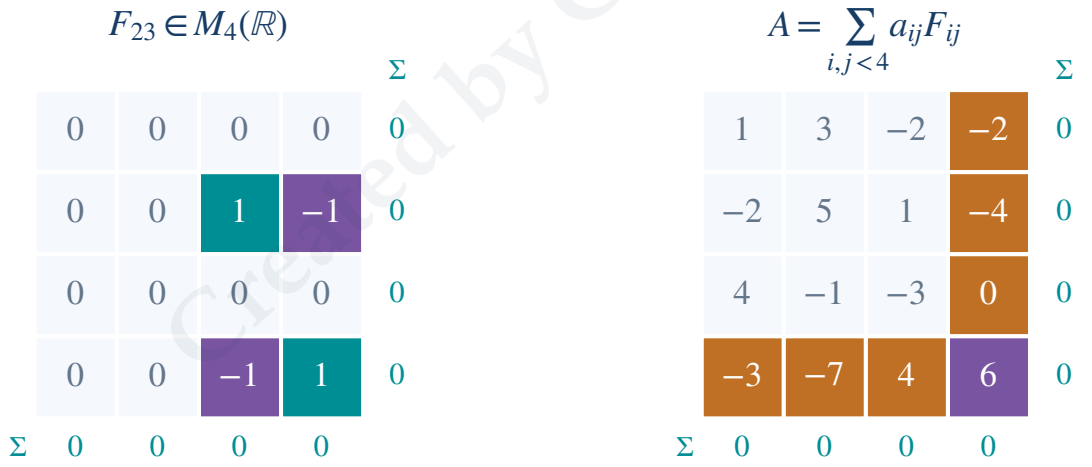


Figure 7.1. Left: the matrix F_{23} for $n = 4$. Right: an interior block completed so that every row and column sum is zero.

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EXERCISE 7.24 CHALLENGE

A permutation matrix has exactly one entry equal to 1 in each row and each column, with all other entries zero. Let $n \geq 2$. Please prove that $A \in M_n(\mathbb{R})$ is a linear combination of permutation matrices if and only if all its row sums and column sums equal the same number.

WORKED SOLUTION

Every permutation matrix has row and column sums equal to 1. Thus every linear combination has all those sums equal to the sum of its coefficients.
Conversely, let c be the common sum of A and put $B = A - cI_n$. All row and column sums of B are zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn},$$

where E_{ij} has one 1 in position (i, j) and zeros elsewhere. Then

$$B = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} b_{ij} F_{ij}.$$

The upper left blocks agree, and zero row and column sums determine the remaining entries. Each F_{ij} is the difference of two permutation matrices. Choose a permutation matrix P with ones in positions (i, j) and (n, n), matching the remaining $n - 2$ rows and columns in any order. Swap rows i, n to obtain a permutation matrix Q. Then

$$P - Q = E_{ij} + E_{nn} - E_{in} - E_{nj} = F_{ij}.$$

Thus every F_{ij} is a linear combination of permutation matrices. The identity is itself a permutation matrix, so $A = B + cI_n$ is a linear combination of them as well.

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